


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

Test Pattern

JEE(Advanced)
AIOT
12-05-2024
JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

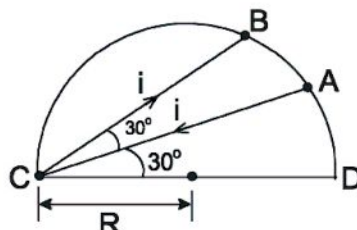
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

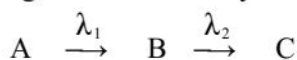
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A current carrying wire is placed along ACB in the grooves of an insulating semicircular disc of radius 'R' as shown. The current enters at point A and leaves at point B. Determine the magnetic field at point D.



- (A) $\frac{\mu_0 i}{8\pi R\sqrt{3}}$
- (B) $\frac{\mu_0 i}{4\pi R\sqrt{3}}$
- (C) $\frac{\sqrt{3}\mu_0 i}{4\pi R}$
- (D) None
2. In the following radioactive decay C is stable nucleus. Then

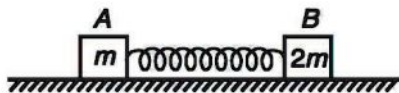


time $t = 0$ N_0 0 0

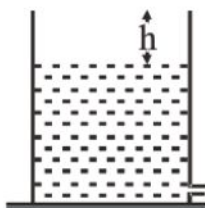
time t N_1 N_2 N_3

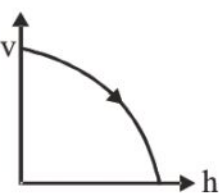
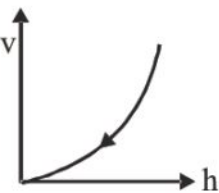
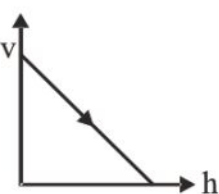
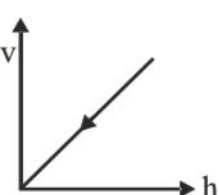
- (A) Number of nuclei of B will first increase and then decrease.
- (B) Rate of decay of A will first increase and then decrease.
- (C) If $\lambda_2 > \lambda_1$ then activity of B will always be higher than activity of A.
- (D) If $\lambda_1 > \lambda_2$ then number of nuclei of C will always be less than number of nuclei of B.

3. In the given figure mass of block A is m and that of B is $2m$. They are attached at the ends of a spring and kept on a smooth horizontal surface. Now, the spring is compressed by x_0 and released. Find displacement of block B by the time compression of the spring is reduced to $\frac{x_0}{2}$:-



- (A) $\left(\frac{x_0}{3}\right)$ (B) $\left(\frac{x_0}{2}\right)$
 (C) $\left(\frac{x_0}{6}\right)$ (D) $\left(\frac{x_0}{4}\right)$
4. A rectangular tank is filled completely with water. A hole at its bottom is unplugged. The graph between the velocity of efflux (through a small hole) vs depth of water h from the top of tank is :-



- (A) 
- (B) 
- (C) 
- (D) 

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

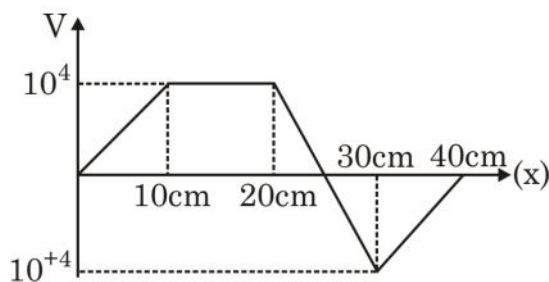
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

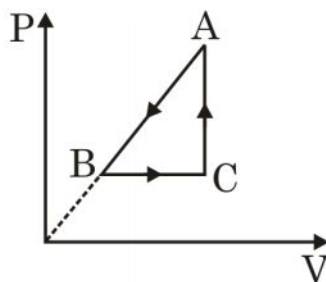
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

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5. The main scale of a vernier calipers reads in mm and its vernier is divided into 10 divisions which coincides with 9 divisions of the main scale. It was also observed when two jaws are brought in contact the zero of the vernier scale coincided with the zero of the main scale. The length of a cube is measured with this vernier calipers. The main scale reads 10 mm and the first division of the vernier scale coincides with the main scale. If the mass of the cube is 2.0606 g then mark the correct alternatives.
- (A) The density of the cube in appropriate significant figures is 2.0000 gm/cm^3 .
- (B) The density of the cube in appropriate significant figures is 2.00 g/cm^3 .
- (C) If the zero mark of the vernier scale lies towards the left side of the zero of the main scale when the two jaws are brought in contact, the error will be positive and the correction will be negative.
- (D) If the zero mark of the vernier scale lies towards the right side of the zero of the main scale when two jaws are brought in contact the error will be positive and the correction will be negative.

6. For the given potential versus distance graph for one dimensional electric field and a particle of charge $20 \mu\text{C}$ and mass 0.2 kg , Choose the correct option(s).



- (A) Force acting on the particle at $x = 25 \text{ cm}$ is zero.
- (B) Force acting on the particle at $x = 25 \text{ cm}$ is $4\hat{i}$ Newton.
- (C) Minimum speed at origin required by the particle to reach $x = 25 \text{ cm}$ is 2 m/s .
- (D) Minimum speed at origin required by the particle to reach $x = 25 \text{ cm}$ is $\sqrt{2} \text{ m/s}$.
7. A sky lab of mass m is first launched from the surface of earth (radius R) in a circular orbit of radius $2R$ (from centre of earth) and then it is shifted from this circular orbit to another circular orbit of radius $3R$.
- (A) Minimum energy required to place the lab in first orbit is equal to $\frac{3}{4} mgR$.
- (B) Minimum energy required to shift the lab from first orbit to second orbit is $\frac{mgR}{12}$.
- (C) Energy of sky lab in second orbit is greater than its energy in first orbit.
- (D) Energy of sky lab in second orbit is less than its energy in first orbit.
8. P-V diagram of cyclic process ABCA is as shown in figure. Choose the correct statement(s)



- (A) $\Delta Q_{A \rightarrow B} = +ve$
- (B) $\Delta U_{B \rightarrow C} = +ve$
- (C) $\Delta U_{CA} = -ve$
- (D) $\Delta W_{CAB} = -ve$

9. A ball of mass m moving horizontally at a speed v collides with the bob of a simple pendulum at rest. The mass of the bob is also m . Here $v \leq \sqrt{2g\ell}$. (Where ℓ is length of the pendulum)
- (A) If the collision is perfectly inelastic, the height to which the two balls rise after the collision is $\frac{v^2}{8g}$.
- (B) If the collision is perfectly inelastic, the kinetic energy of the system immediately after the collision becomes half of that before collision.
- (C) If the collision is perfectly elastic, the bob of the pendulum will rise to a height of $\frac{v^2}{2g}$.
- (D) If the collision is perfectly elastic, the kinetic energy of the system immediately after the collision is equal to that before collision.
10. A pulley of radius $R = 20$ cm is rotated by a weight suspended from a thread which is gradually being wound off the pulley. Initially the weight is at rest and then it begins to go down with an acceleration $a = 2 \text{ cm/s}^2$:-



- (A) Angular velocity of the pulley when the weight has travelled a distance $S_1 = 100$ cm is 1 rad/s .
- (B) Angular velocity of the pulley when the weight has travelled a distance $S_1 = 100$ cm is 2 rad/s .
- (C) The magnitude of the acceleration of point A when the weight has travelled a distance $S_1 = 100$ cm is $20\sqrt{2} \text{ cm/s}^2$.
- (D) The magnitude of the acceleration of point A when the weight has travelled a distance $S_1 = 100$ cm is 20.1 cm/s^2 .

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

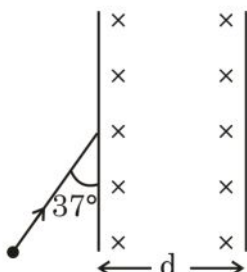
Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. Two thin lenses with lens powers D_1 and D_2 diopter are located at distance $L = 25$ cm from each other and their optical axes coincide. This system creates an erect real image of the object, located at the optical axis closer to lens D_1 with the net magnification $M_1 = 1$. If the position of the two lenses are exchanged, the system again produces an erect real image with the net magnification $M_2 = 4$. What is the difference between lens powers $\Delta D = D_1 - D_2$ (in diopter)?
2. A storage battery ($E = 24$ V) was connected to one end of a two conductor telephone line of length $L = 5.6$ km. There is a leakage somewhere in the line. If the free ends are open, then current through the battery is $I_1 = 1.6$ A and if the free ends of the line are short-circuited then current through the battery is $I_2 = 2$ A. The short-circuit current of the battery is $I_3 = 24$ A and resistance of each conductor is $r = 7\Omega$. Determine the resistance (in Ω) of the insulation R at the point of the breakdown.
3. A string will break apart if it is placed under too much tensile stress. One type of steel has density $\rho_{\text{steel}} = 10^4 \text{ kg/m}^3$ and breaking stress $\sigma = 9 \times 10^8 \text{ N/m}^2$. We make a guitar string from (4π) gram of this type of steel. It should be able to withstand (900π) N without breaking. The highest possible fundamental frequency (in Hz) of standing waves on the string is $75n$. Find n .
4. A car begins to move at time $t = 0$ and then accelerates along a straight track with a velocity given by $V(t) = 2t^2 \text{ ms}^{-1}$ for $0 \leq t \leq 2$, where t is time in second. After the end of acceleration, the car continues to move at a constant speed. A small block initially at rest on the floor of the car begins to slip at $t = 1$ s and stops slipping at $t = 3$ s. The coefficient of static and kinetic friction between the block and the floor are μ_s and μ_k respectively. Find the value of $\frac{3\mu_s}{\mu_k}$.
5. A plane spiral with a great number N of turns wound tightly to one another is located in a uniform magnetic field perpendicular to the spiral's plane. The outside radius of the spiral's turns is equal to a and inner radius is zero. The magnetic induction varies with time as $B = B_0 \sin \omega t$, where B_0 and ω are constants. Find the amplitude of emf induced in the spiral. (Given : $\omega = \frac{1}{\pi} \text{ rad/s}$, $N = 1000$, $a = 3\text{cm}$, $B_0 = 10 \text{ Tesla}$)

6. A steel rod is 4.000 cm in diameter at 30°C. A brass ring has an interior diameter of 3.992 cm at 30°C. In order that the ring just slides onto the steel rod, the common temperature of the two should be θ (in °C). Find $\theta/100$ rounded off to nearest integer ($\alpha_{\text{steel}} = 11 \times 10^{-6}/^{\circ}\text{C}$ and $\alpha_{\text{brass}} = 19 \times 10^{-6}/^{\circ}\text{C}$)
7. A particle of mass 1 mg and charge $1\mu\text{C}$ is projected with speed 50 m/s towards a region containing a uniform magnetic field 10 T. What should be the minimum thickness (d) of this region (in m) so that the particle returns back to left side?



8. A point charge is moving along a straight line with constant velocity v . Consider a small area A perpendicular to the direction of motion of charge as shown in figure.



The displacement current through the area when its distance from the charge is x is $\frac{qAv}{2\pi x^n}$. The value of n is :-

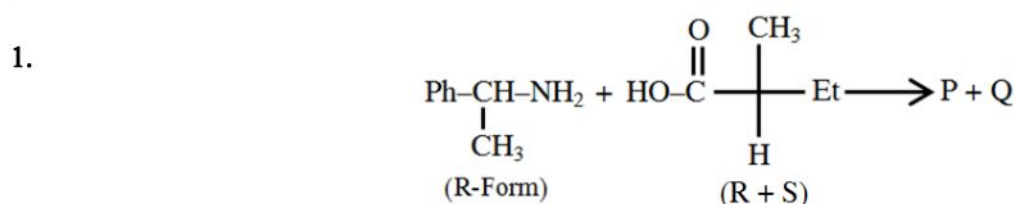
PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

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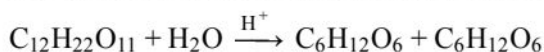
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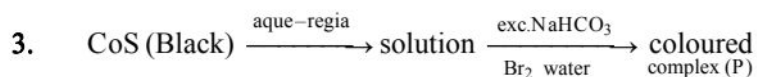
The correct statement regarding product P & Q is :

- (A) Both are optically active esters
 - (B) P & Q are diastereomers
 - (C) P & Q are meso compounds
 - (D) Out of P & Q, one is optically active and other is optically inactive.
2. Inversion of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is first-order reaction and is studied by measuring angle of rotation of the reaction mixture at different instants of time



If r_0 , r_t and r_∞ are the angles of rotation of the reaction mixture at the start, at the time t and at the end of the reaction respectively, then 50% inversion of sucrose occurs when :

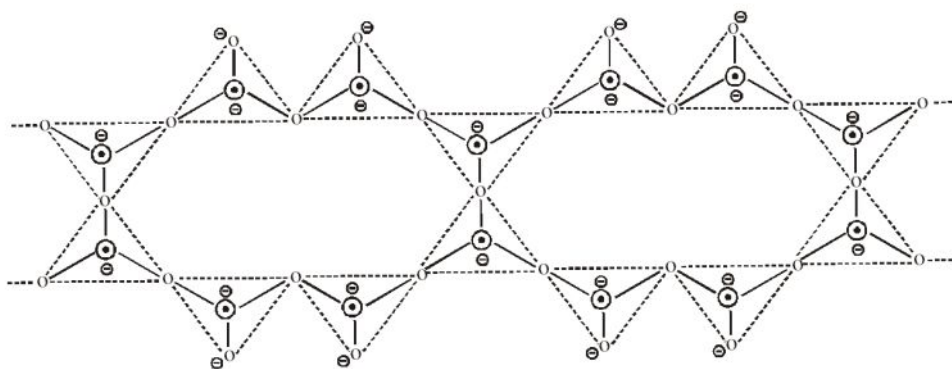
- (A) $r_0 = 2r_t - r_\infty$
- (B) $r_0 = r_t - r_\infty$
- (C) $r_0 = r_t - 2r_\infty$
- (D) $r_0 = r_t + r_\infty$



What is co-ordination no. of central metal in complex (P)

- (A) 5
- (B) 6
- (C) 3
- (D) 4

4. What is general formul of below mentioned silicate ? [o : oxygen; • : silicon]



- (A) $(\text{Si}_4\text{O}_{11}^{6-})_n$
- (B) $(\text{Si}_2\text{O}_5^{2-})_n$
- (C) $(\text{Si}_6\text{O}_{17}^{10-})_n$
- (D) $(\text{Si}_3\text{O}_8^{4-})_n$

SECTION-I (ii) : (Maximum Marks: 24)

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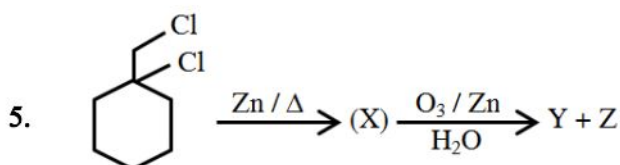
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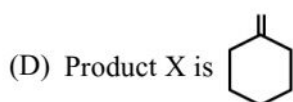
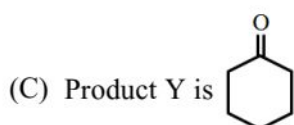
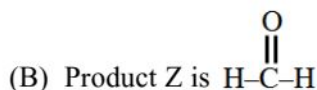
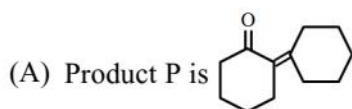
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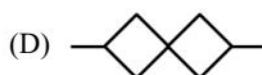
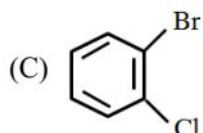
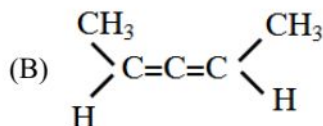
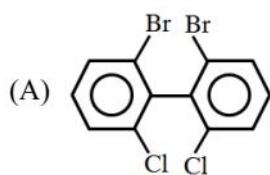
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Product Y and Z are separated and then heated with NaOH. Y gave P where as Z gave Q and CH_3OH as product then which of the following is/are correct :



6. Which of the following is/are resolvable ?



7. Select the correct statement(s) from the following.

(A) If the critical micelle concentration (CMC) of a soap is 10^{-3} M, then 10^{-4} M solution of this soap will have colloidal nature

(B) Charge on the colloidal particles may be determined with the help of dialysis.

(C) Peptization is the process of conversion of a colloidal sol into precipitate by adding some electrolyte.

(D) When some oil-soluble dye is added in the water in oil emulsion, the background becomes coloured.

8. EMF of the cell: $\text{Cd(s)} \mid \text{CdCl}_2 \cdot 5\text{H}_2\text{O (sat.)} \mid \text{AgCl(s)} \mid \text{Ag(s)}$ is +0.70 V at 0°C and +0.60 V at 50°C . If ΔH° and ΔS° are temperature independent, then the correct information(s) regarding the cell reaction is/are

(A) $\Delta G^\circ = -115.8$ kJ at 50°C

(B) $\Delta G^\circ = 135.1$ kJ at 0°C

(C) $\Delta S^\circ = -386$ J/K

(D) $\Delta H^\circ = -221.178$ kJ

9. Select the correct comparison(s) of properties indicated in options :
- (A) $\text{MnO}_4^{2-} > \text{MnO}_4^-$: (paramagnetic character)
 - (B) $\text{Na}_2\text{Cr}_2\text{O}_7 > \text{K}_2\text{Cr}_2\text{O}_7$ (Solubility in water)
 - (C) $\text{Lu}^{+3} < \text{Ce}^{+3} < \text{La}^{+3}$: (Ionic radius)
 - (D) $\text{Mn}^{+3} < \text{Fe}^{+3} < \text{Ti}^{+3}$: (oxidising power)
10. The extraction of pure metal from bauxite ore (containing iron oxide as chief impurity) involves :
- (A) Electrolytic reduction
 - (B) Poling
 - (C) Leaching by exc. NaOH(aq.)
 - (D) Precipitation of amphoteric hydroxide

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SECTION-III : (Maximum Marks: 24)

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- For each question, marks will be awarded in one of the following categories :

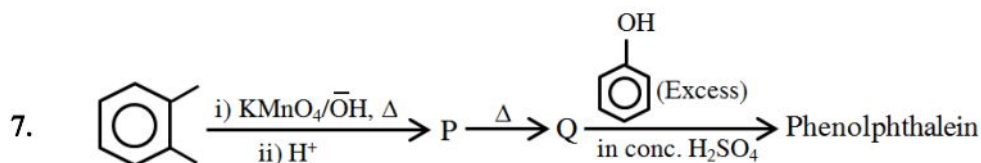
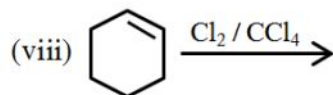
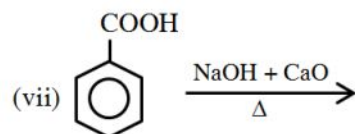
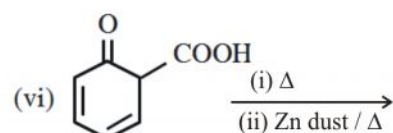
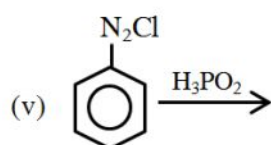
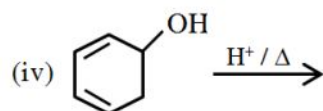
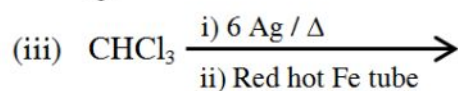
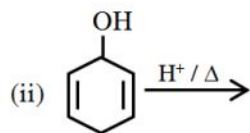
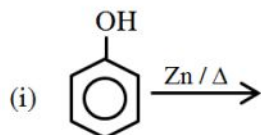
Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

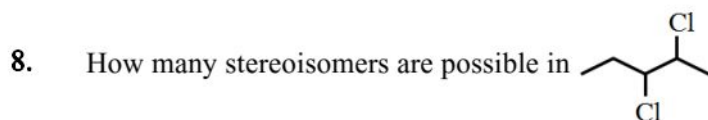
Negative Marks : -1 In all other cases

1. A container of capacity 20 L contains $N_2(g)$, water vapour and negligible but sufficient amount of liquid water to maintain equilibrium, at 300K and 1 atm. The liquid water is now decomposed completely into $H_2(g)$ and $O_2(g)$, by any means, without changing the temperature. If the final pressure becomes 1.9 atm, the mass of liquid water present initially (in gm) was (Nearest integer)
(Aqueous tension of H_2O at 300K = 0.04 atm, $R = 0.08$ L-atm/K-mole)
2. Cd^{2+} ion is precipitate after passing H_2S in acidified solution. In the final solution 0.1 milli-mole of $CdSO_4$ are present in 10 ml acidic saturated solution of 0.10M H_2S . What is the pH of the solution after filtering off precipitate, boiling off H_2S and making the solution 100 ml by adding H_2O :
[K_{sp} of $CdS = 10^{-28}$, For H_2S : $K_{a1} = 10^{-9}$, $K_{a2} = 10^{-20}$]
3. Let us assume that de-Broglie equation is valid in Bohr model. In H-like sample, an electron is in an orbit which has path length of the orbit is equal to 3 times of wavelength of electrons in that orbit. This orbit of H-like sample has total energy equal to the ground state energy of H-atom. Now we have same sample having all electrons in the ground state. If photon of energy 126.566 eV is given to this ground state sample then approximate value of wavelength (λ)(in Å, closest integer) of ejected e^- is :
4. Iodine reacts with chlorine in presence of water to give an iodine containing compound 'x'. How many moles of 'x' is formed on reaction of 2 moles I_2 with 10 moles of Cl_2
5. A solution contains equimolar amount of following ions : Zn^{+2} , Fe^{+3} , Ag^+ , Cu^{+2} , Ni^{+2} , Pb^{+2} , Cd^{+2} , Hg^{+2} , Cr^{+3} . How many of cations form soluble complex with excess of aqueous NH_3 ?

6. How many of the following reactions can give Benzene as final product.



Find out degree of unsaturation of product Q :



PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. If x_1 and x_2 are the roots of the equation $\log_{\pi}((\sin^{-1}(x))(\cos^{-1}(x))) = \log_{\sqrt{\pi}}\left(\frac{\pi}{\sqrt{18}}\right)$ for $x \in (0, 1)$ then the value of $\tan^{-1}(x_1^2 + x_2^2)$ is equal to -
- (A) $\frac{\pi}{3}$
(B) $\frac{\pi}{4}$
(C) $\frac{\pi}{6}$
(D) $\frac{\pi}{8}$
2. Let a_n and b_n be the sequences of real numbers such that for all integers $n \geq 0$ where $i = \sqrt{-1}$ and $(3 + i)^n = (a_n + b_n i)$ then the value of $\sum_{n=0}^{\infty} \frac{a_n b_n}{12^n}$ is equal to
- (A) $\frac{9}{13}$
(B) $\frac{5}{13}$
(C) $\frac{12}{13}$
(D) $\frac{7}{13}$

3. $\int_0^{\pi/3} \frac{\{\tan x\}}{\tan x} dx$ is equal to ($\{x\}$ denotes fractional part of x)

(A) $\frac{\pi}{3} + \frac{1}{2} \ln \left(\frac{3}{2} \right)$

(B) $\frac{\pi}{3} - \frac{1}{2} \ln \left(\frac{3}{4} \right)$

(C) $\frac{\pi}{3} - \frac{1}{2} \ln \left(\frac{3}{2} \right)$

(D) $\frac{\pi}{3} + \frac{1}{2} \ln \left(\frac{3}{4} \right)$

4. In a cube ABCDEFGH with opposite vertices C and E, J and I are the mid points of segments FB and HD respectively. Let R be the ratio of the area of the cross section EJCI to the area of one of the faces of the cube then value of R^2 is equal to -

(A) $\frac{5}{4}$

(B) $\frac{3}{2}$

(C) $\frac{4}{3}$

(D) $\frac{25}{16}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. If ${}^nC_2 = f(n)$ and $S = {}^{f(3)}C_2 + {}^{f(4)}C_2 + {}^{f(5)}C_2 + \dots + {}^{f(40)}C_2$ then which options are correct for S

- (A) Total number of divisors are 144
- (B) Total number of odd divisors are 48
- (C) S is divisible by 78
- (D) Total number of prime factors are 7

6. Let z be a complex number satisfying $12|z|^2 = 2|z + 2|^2 + |z^2 + 1|^2 + 31$ then which options are correct
- (A) $\left|z + \frac{6}{z}\right| = 2$
- (B) $\operatorname{Re}(z) = -1$
- (C) $|z| = \sqrt{6}$
- (D) $\frac{3\pi}{4} < \operatorname{Arg}(z) < \pi$
7. For regular tetrahedron ABCD, points E, F are on AB, AC respectively, such that $BE = 3$, $EF = 4$, and EF is parallel to plane BCD. Then which options are correct
- (A) $DE = \sqrt{37}$
- (B) Area of $\triangle DEF = 2\sqrt{33}$
- (C) Height of tetrahedron $\sqrt{33}$
- (D) Volume of tetrahedron $= \frac{49}{4} \sqrt{\frac{37}{3}}$
8. For triangle ABC, $A(0, 8)$, $B(6, 0)$ and $C(x, y)$ are three points such that $C(x, y)$ lie on circle $(x - 8)^2 + (y - 6)^2 = 9$. If Δ is the area of triangle ABC and z_m & z_M is a point corresponding to complex number of point C for which area is minimum and maximum respectively then select correct options
- (A) $\Delta_{\max} = 41$
- (B) $\Delta_{\min} = 11$
- (C) $\arg\left(\frac{5(z_M + \bar{z}_M) - 100}{5(z_M - \bar{z}_M) - 78}\right) = -\frac{\pi}{2}$
- (D) $\arg(z_m) < \frac{\pi}{4}$

9. Let $f : \mathbb{N} \rightarrow \mathbb{Q}$ be a function where $\mathbb{N}$ denotes the set of natural numbers and $\mathbb{Q}$ denotes the set of rational numbers. $f(1) = \frac{3}{2}$ and $f(x+y) = \left(1 + \frac{y}{x+1}\right) f(x) + \left(1 + \frac{x}{y+1}\right) f(y) + x^2y + xy + xy^2$ for all natural numbers x, y the select correct options :-
- (A) $f(20) = 4305$
- (B) If $f(n) = 1$ then only one root lie only between $(0, 1)$
- (C) $3^2 + 6^2 + 9^2 + \dots + (3n)^2 = 6 f(n)$
- (D) If $2f(n) = (2023) {}^{(n+1)}C_2$ then $n = 1011$
10. Let $y(x)$ be a solution of the differential equation $\left(x - \frac{dy}{dx}\right) \sin x = y \cos x$, $x \in \left(0, \frac{\pi}{2}\right]$. If $y\left(\frac{\pi}{2}\right) = 1$ then choose the correct statement(s) :
- (A) $y(x)$ attains a maximum value in $\left(0, \frac{\pi}{2}\right]$
- (B) $y(x)$ attains a minimum value in $\left(0, \frac{\pi}{2}\right]$
- (C) $\lim_{x \rightarrow 0^+} y(x) = 0$
- (D) $0 < y\left(\frac{\pi}{4}\right) < 1$

Note : No questions have been asked in the section II of this paper. Please leave section II blank on the OMR answer sheet.

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. For parabola $y^2 = 4x$, let $P(11, b)$ be any point inside the parabola and F be the focus of the parabola. If the point Q on the parabola such that $FQ + QP$ is minimum and R is a point of intersection of axis and directrix then the area of parallelogram $FPQR$ is equal to
2. Let A and B are 2×2 matrice such that $\text{tr}(A) = 7$, $\det(A) = 22$, $\text{tr}(B) = 4$, $\det(B) = 11$, $\text{tr}(AB) = 33$. If $\det(A + xB) = 272$ then the value of x is ($x \in \mathbb{I}^+$)
3. Let n be a positive integer such that exactly one root of the equation $4x^2 - (4\sqrt{3} + 4)x + \sqrt{3}n - 24 = 0$ is an integer. Then the value of $\frac{n}{2}$ is
4. If $I = \int_0^{\infty} \frac{1}{(1+x^{2024})(1+x^2)} dx$ then the value $[I]$ ($[]$ Greatest Integer Function)
5. Let $f: \{1, 2, 3, 4\} \rightarrow \{a, b, c, d, e, f\}$. If $f(1) \neq a$ or $f(2) \neq b$ or both then number of functions are k then sum of digits of k is
6. A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 compononts randomly from a lot and to accept the lot only if all 3 are non defective. If 30% of the lots have x defective components and 70% have only 1. If purchases reject 46% of lots then value of x is
7. Let T_n consists of general term of positive numbers (where n is positive integer). If $\frac{T_n + 2}{2} = \sqrt{2S_n}$ and S_n is sum of n terms, then the value of $(T_{2024} - 8090)$ is equal to
8. Let $f(x) = e^{-|x|}$, $x \in \mathbb{R}$ and $g(t) = \int_{-1}^1 f\left(\frac{x}{t}\right) dx$, $t \neq 0$ then value of $\lim_{t \rightarrow 0^+} \frac{g(t)}{t}$ is equal to

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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